Blockchain : Experiment - 4

Aim:

Creation of account in Metamask, Transfer of Amount/Cryptocurrency from one account to another.

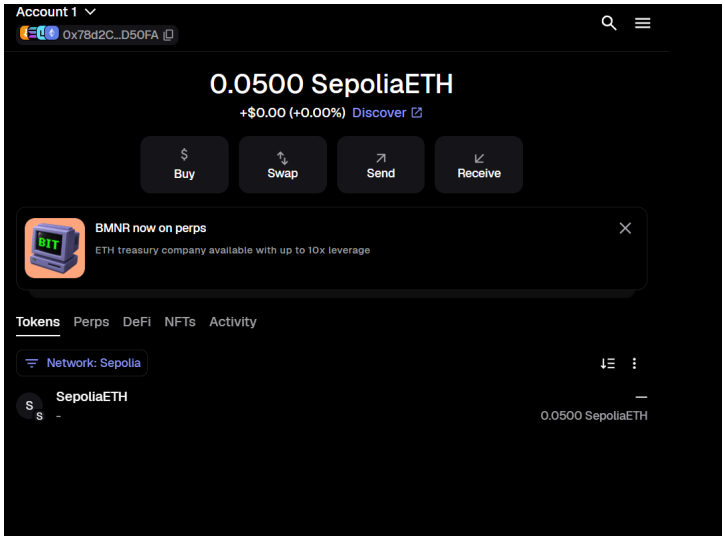
Theory:

1. **What is MetaMask?**
   MetaMask is a non-custodial cryptocurrency wallet, available as a browser extension and mobile app, that lets a user generate and manage Ethereum-compatible accounts, hold ETH and other tokens, and interact with decentralized applications and blockchains directly from the browser. Being non-custodial means the wallet's private keys are generated and stored locally on the user's device (encrypted with a password, and recoverable via a 12-word secret recovery phrase) rather than being held by any company or exchange, so the account holder has full control of, and full responsibility for, their funds.
2. **Account Creation in MetaMask**
   When a new MetaMask account is created, the wallet generates a public/private key pair. The private key is used to sign transactions and must never be shared, while the public key is hashed to derive the account's public wallet address (e.g., 0xbb6E6...189B4) — the identifier others use to send funds to that account. A single MetaMask installation can hold multiple accounts (all derived from the same seed phrase, or imported separately), letting one user manage several addresses from a single interface.
3. **Testnets and Test Tokens**
   Since real ETH has monetary value, blockchain transactions during development and learning are usually performed on a testnet — a separate blockchain that mimics the mainnet's behaviour but uses worthless test tokens. In this experiment, the Sepolia Testnet was used, and the account was funded with test SepoliaETH obtained from a Sepolia faucet, allowing transactions to be sent and confirmed without any real financial cost or risk.
4. **How a Transaction Works**
   A transaction in MetaMask/Ethereum consists of a sender address, a recipient address, an amount to transfer, and a gas fee paid to the network for processing it. When a user initiates a "Send" transaction:
   - The sender specifies the recipient's address and the amount to transfer.
   - MetaMask estimates the network fee (gas) required to process the transaction, based on current network congestion.
   - The user reviews the transaction details — sender, recipient, network, amount, and fee — and confirms it by signing it with their private key.
   - The signed transaction is broadcast to the network, picked up by validators/miners, included in a block, and once mined, the recipient's balance is updated and the sender's balance is reduced by the amount sent plus the gas fee.

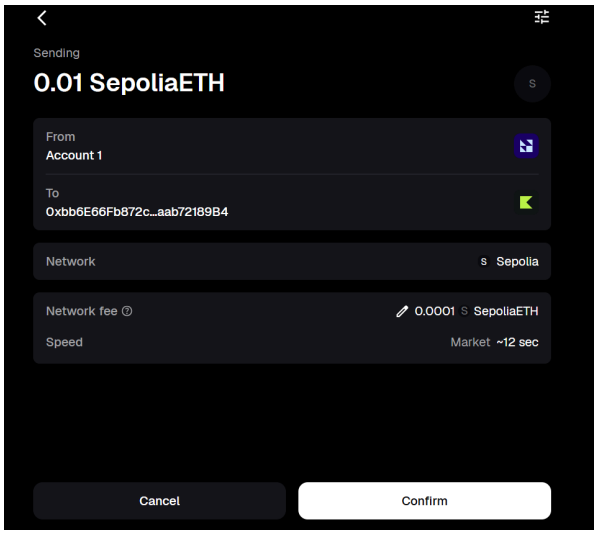
This process demonstrates the core blockchain principle of peer-to-peer value transfer without an intermediary such as a bank: the transaction is validated and recorded directly on the distributed ledger, and is publicly verifiable by anyone via a block explorer.

Output:

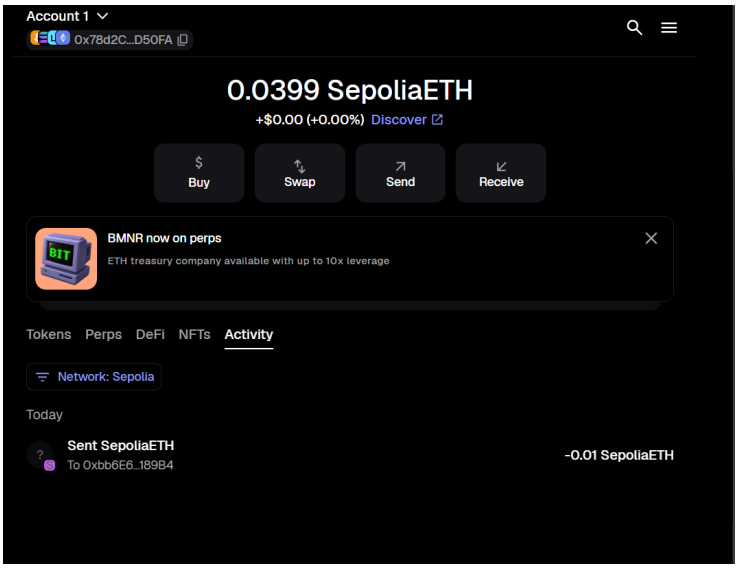
1. **Account created and funded**
   A MetaMask account was set up on the Sepolia Testnet and funded with 0.0500 SepoliaETH from a testnet faucet.

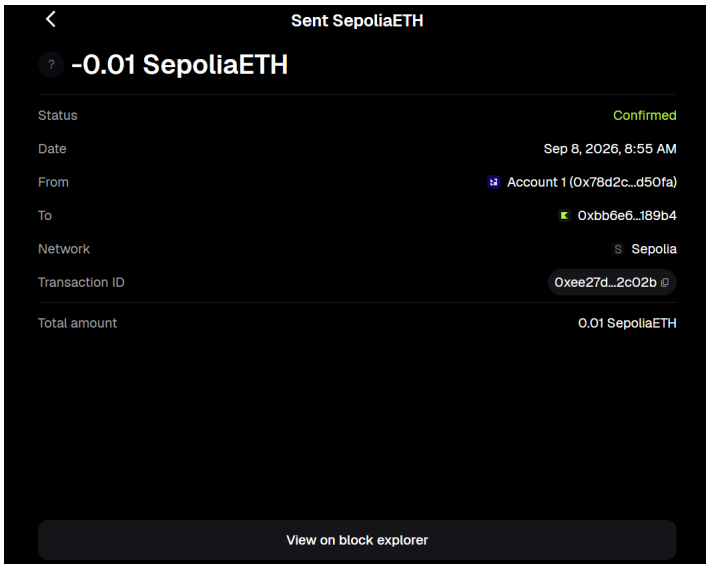


2. **Initiating the transfer**
   The "Send" option was selected, the recipient address was entered, and an amount of 0.01 SepoliaETH was specified for transfer.



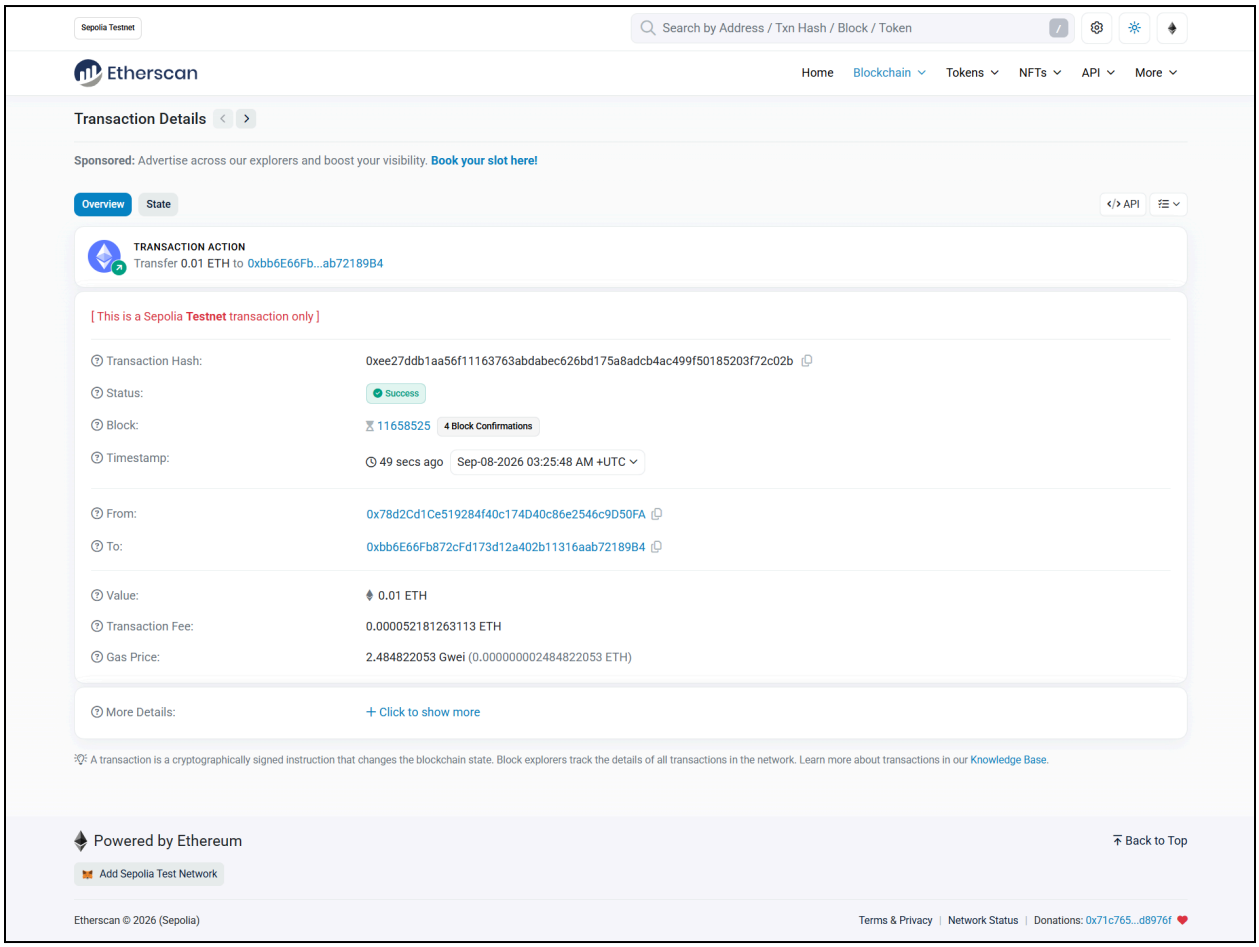
3. **Confirming the transaction**
   MetaMask displayed a confirmation screen summarizing the transaction — sender, recipient address, network (Sepolia), amount (0.01 SepoliaETH), and the estimated network fee (0.0001 SepoliaETH). The transaction was confirmed and signed.





4. **Verifying the updated balance**
   After the transaction was mined, the wallet balance updated to 0.0399 SepoliaETH — reflecting the 0.01 ETH sent plus the gas fee deducted from the original 0.0500 SepoliaETH balance, confirming the transfer was successfully completed.

Transaction On Block Explorer:



Conclusion:

In this experiment, a MetaMask wallet account was created and funded with test SepoliaETH on the Sepolia Testnet. A peer-to-peer transaction of 0.01 SepoliaETH was initiated, reviewed, and confirmed through MetaMask's Send flow, and the resulting change in account balance (from 0.0600 to 0.0499 SepoliaETH, accounting for the amount sent and the network gas fee) confirmed that the transaction was successfully broadcast and mined on the network. This experiment demonstrated the fundamental blockchain workflow of wallet-based, peer-to-peer value transfer without any intermediary, and illustrated how gas fees are deducted alongside the transferred amount in every on-chain transaction.